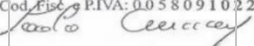


UDROŻNIENIE ŁÓDZKIEGO WĘZŁA KOLEJOWEGO (TEN-T), ETAP II, ODCINEK ŁÓDŹ FABRYCZNA - ŁÓDŹ KALISKA/ŁÓDŹ ŻABIENIEC – PROJEKT POIIS 5.1-15

TBM 13.08 M - CONSTRUCTION FOLLOW-UP

IDENTIFICATION TABLE

Client	PBDiM - Energopol
Project	Udrożnienie Łódzkiego Węzła Kolejowego (TEN-T), Etap II, Odcinek Łódź Fabryczna - Łódź Kaliska/Łódź Żabieniec – Projekt POIIS 5.1-15
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1. INTRODUCTION

S-1222 TBM EPB machine with 13.08 m excavation diameter is currently mining Axis 23, in the section between Polesie Station and Śródmieście Station. TBM has restarted from chainage 1+962.

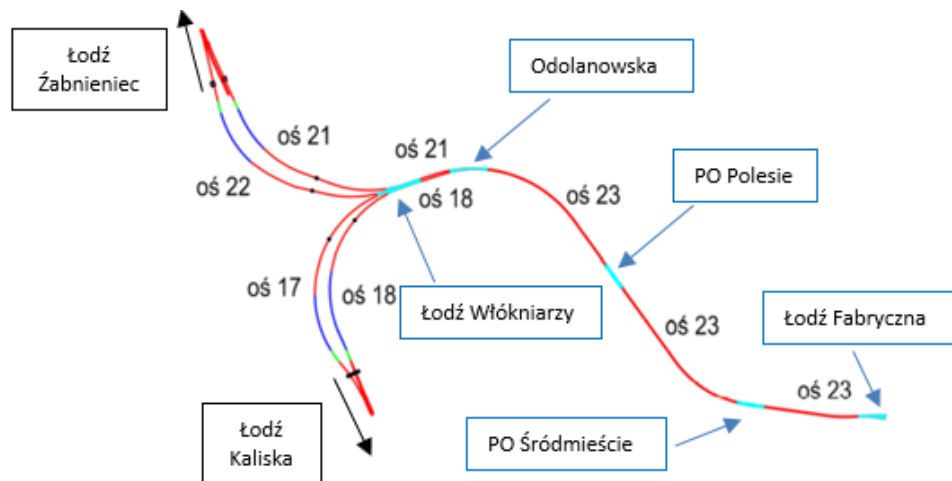


Figure 1-1 Overview of main structural elements of the Project

This daily follow-up report aims to give a general overview of S-1222 TBM drive during 04/09/2024. It includes EPB face pressure, belt scale parameters, thrust and torque values, primary grout parameters, guidance, as well as monitoring follow-up data.

According to VMT system, TBM is at Chainage 1+647.16 after the erection of ring n. 258; which corresponds to a total excavation length of 319.31 m from the starting chainage.

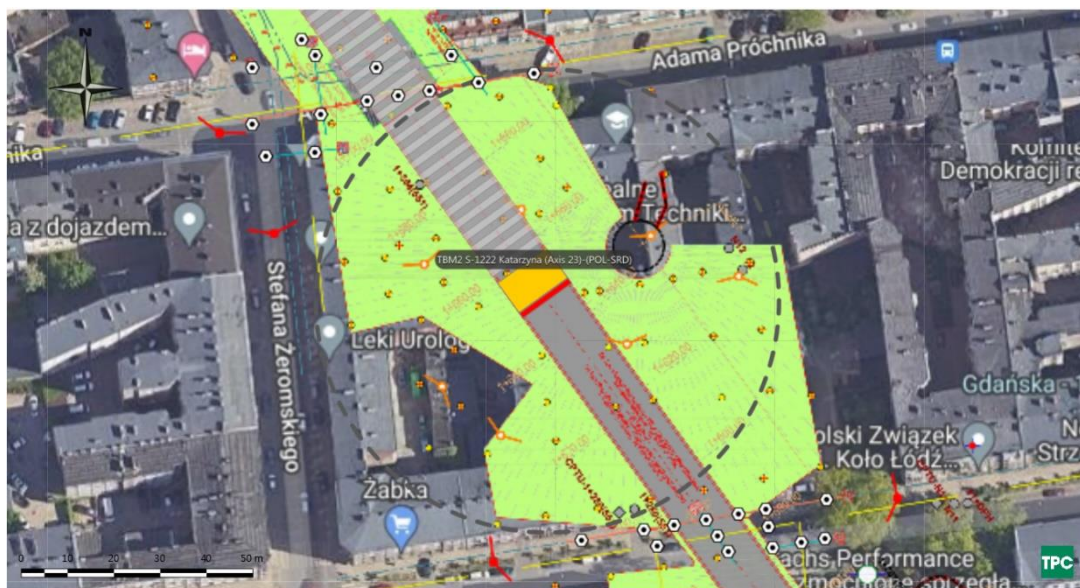


Figure 1-2 Plan view of TBM Position on 05/09/2024

2. FACE PRESSURES AND EXCAVATION VOLUMES

Average EPB pressure at the crown is at about 1.7 bar. It was noted that a drop in pressures below minimum design pressures occurred at 2:00 am, due to attempts to unblock the shield and to advance after the maintenance stop. The values are then recovered within the attention values.

Figure 2-1 shows EPB sensors.

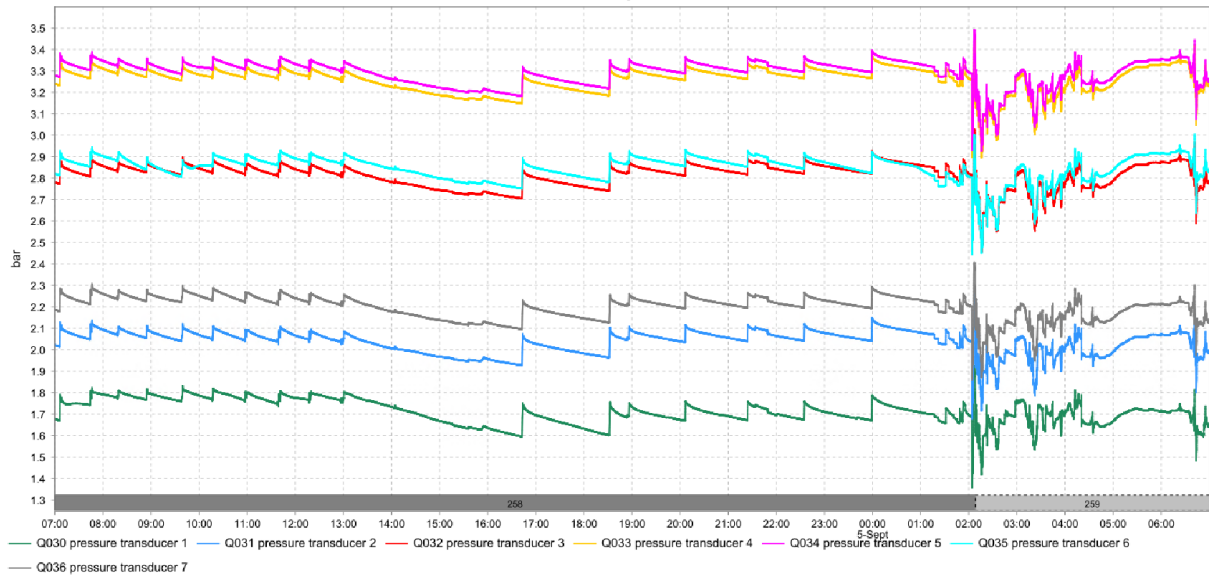


Figure 2-1 EPB pressure sensor

The delta between pressure transducer 1 and 2 has an average value of 0.32 bar indicating that the bulk chamber is correctly filled with material.

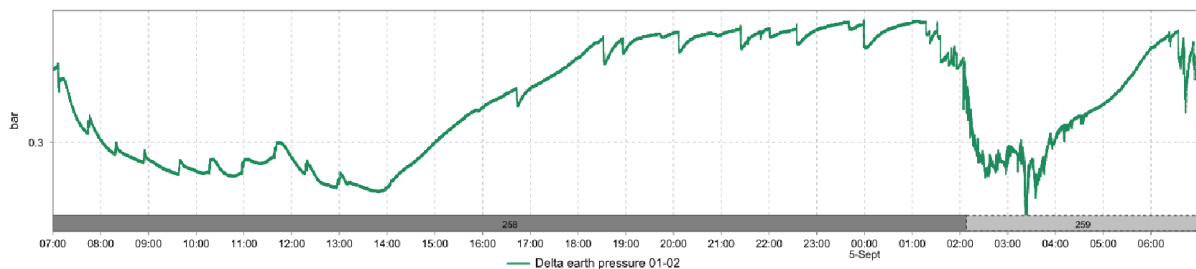


Figure 2-2 Delta between pressure sensors 1 and 2

Belt scale n. 2 show values lower than n. 1, but closer to each other, for ring 258. The belt scale value is a raw indication of the excavated material quantity.

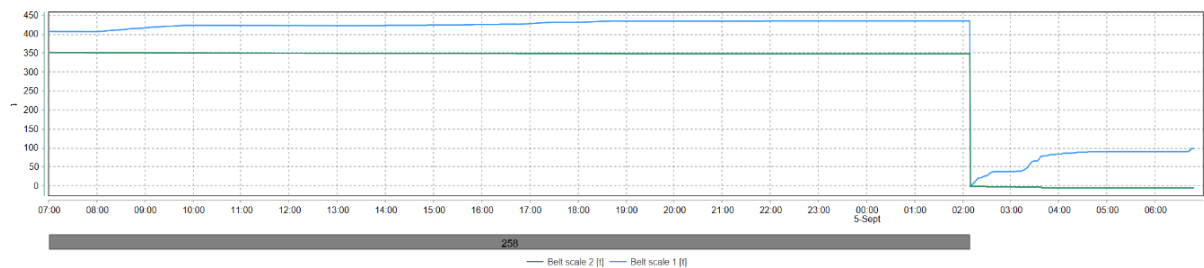


Figure 2-3 Belt scales

Table 2-1 Belt scale values at the end of each advance

Ring	Belt scale 1 (t)	Belt scale 2 (t)
258	436	353

3. THRUST AND CUTTERHEAD TORQUE

During excavation, the average torque is of 1.1 MNm.

During advancement, average advance rate is of 0.5 mm/min.

During excavation, thrust force shows an average value of 79.5 MN and a maximum of 88.6 MN, due to attempts to unblock the shield and to advance after the maintenance stop.

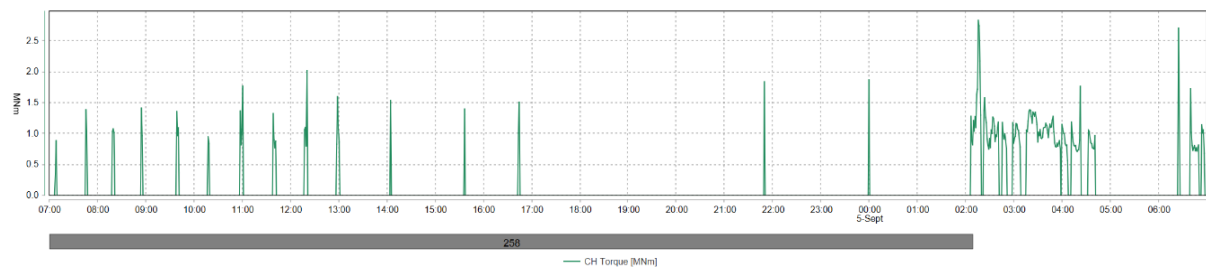


Figure 3-1 CH Torque

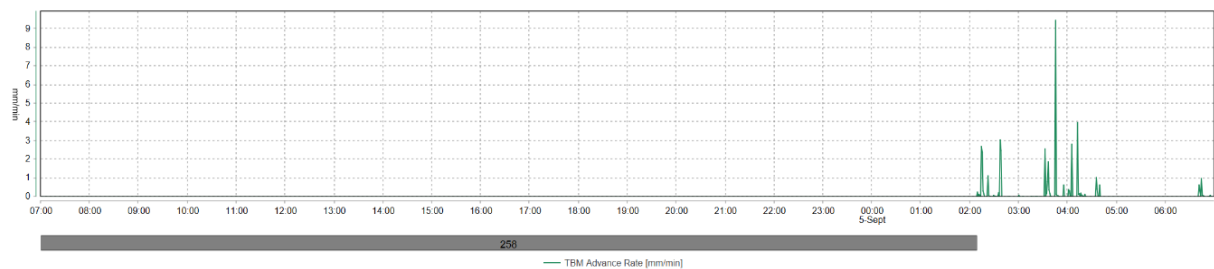


Figure 3-2 Advance rate

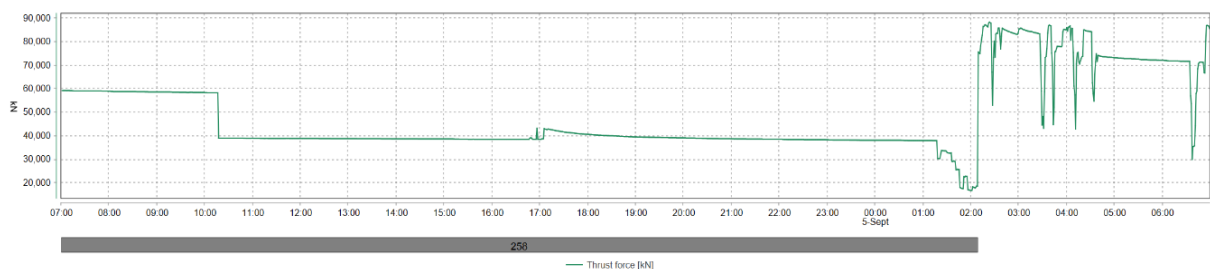


Figure 3-3 Thrust force

4. GROUT INJECTION

Grout injection has a volume above the minimum theoretical annular volume $\pm 10\%$ of 13.54 m^3 . Ratio B/A is at 4% .

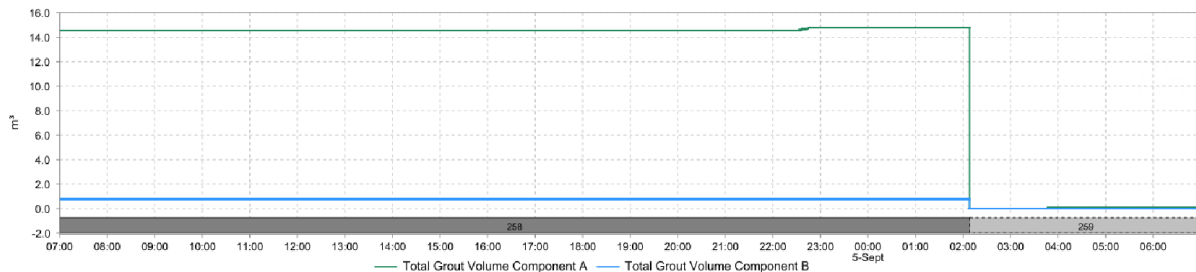


Figure 4-1 Grout volumes

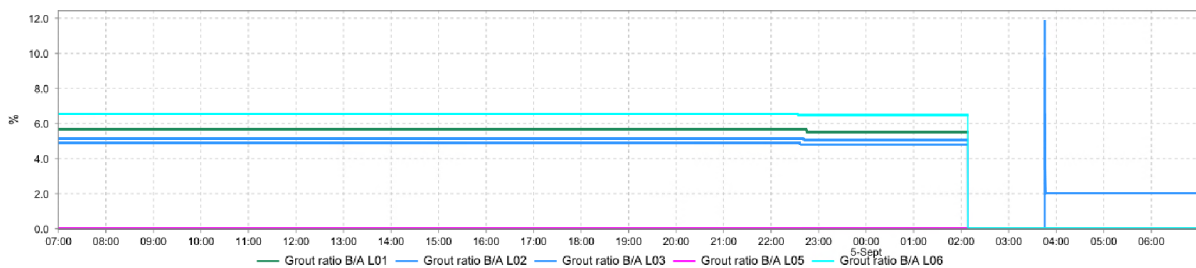


Figure 4-2 Volume ratio Components B/A

Table 4-1 Grout volumes per ring

Ring	Component A (m^3)	Component B (m^3)	B/A (%)
258	14.8	0.8	5

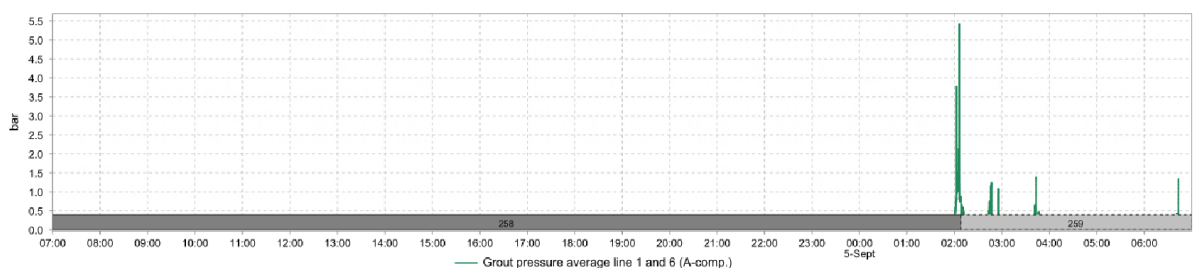


Figure 4-3 Average injection pressure line 1 and 6

5. GUIDANCE

The TBM vertical deviation is of about -2.03 mm on the front edge of the shield and the horizontal deviation of about -3.61 mm (Figure 5-1). Maximum radial deviation is of 7.57 mm.

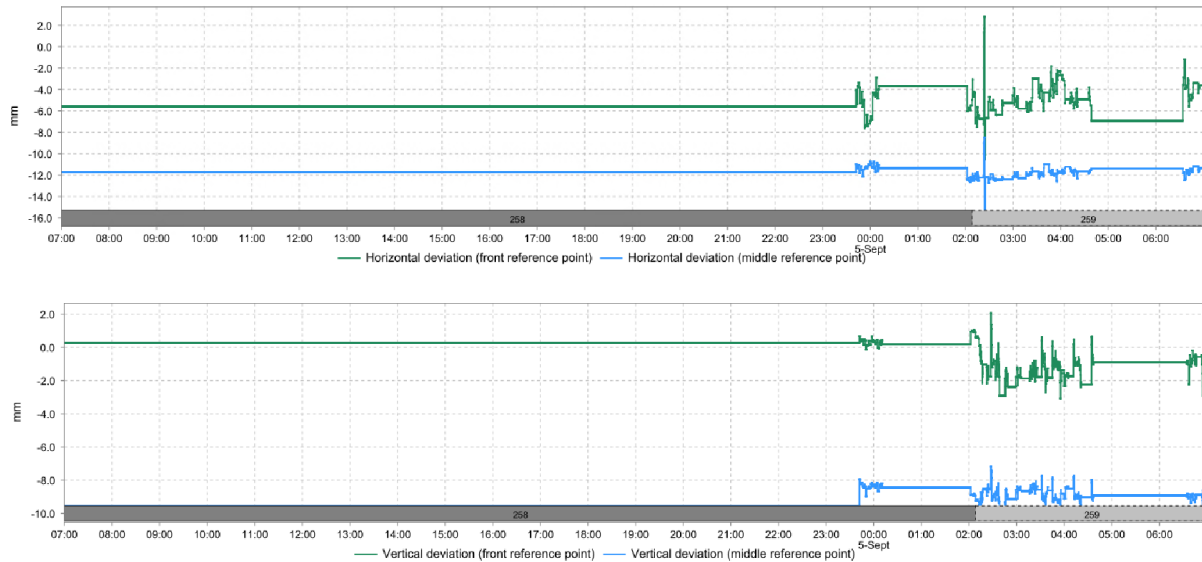


Figure 5-1 Machine deviation from the alignment

6. MONITORING

According to Geoscope, settlements of the buildings above the tunnel are kept within the limits.



Figure 6-1 Monitoring points

On PR0440, monitoring points show a small uplift trend for points: PR0440_3R and PR0440_05R.

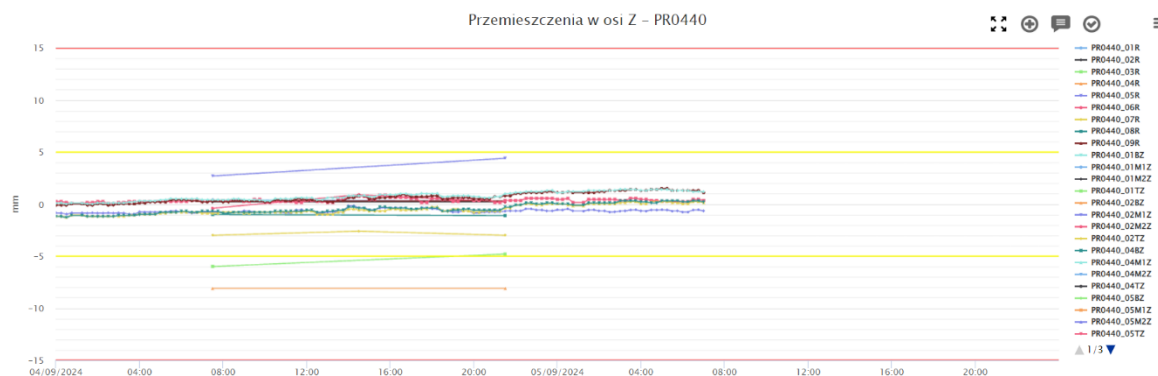


Figure 6-2 Settlement of building PR0440

On PR0460, monitoring points are stable.

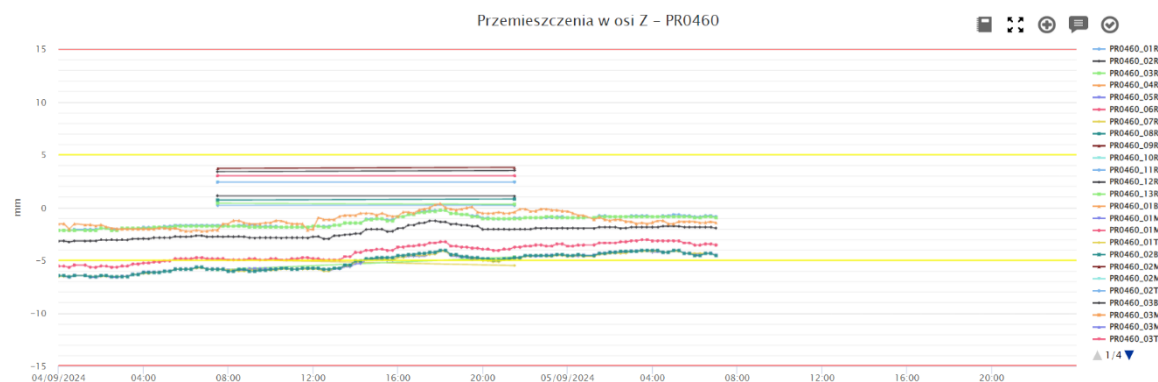


Figure 6-3 Settlement of building PR0460

On MA0190, monitoring points are stable.

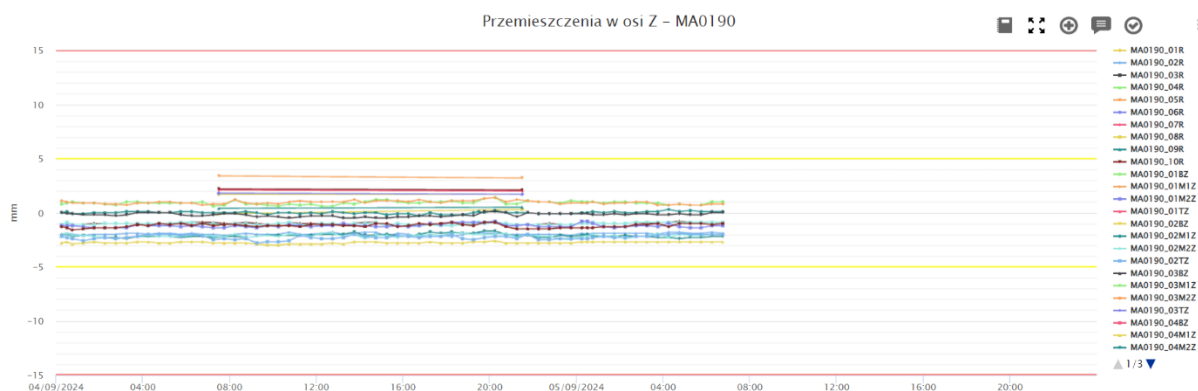


Figure 6-4 Settlement of building MA0190

On MA0210, monitoring points are stable.

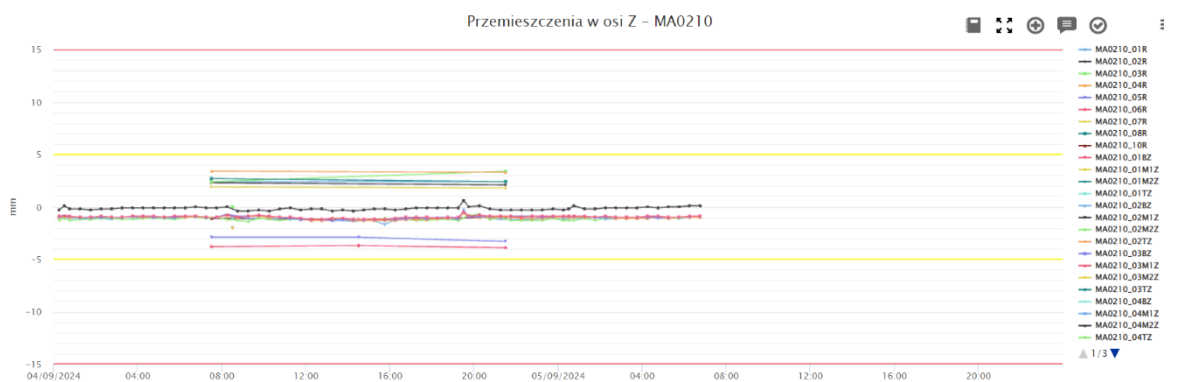


Figure 6-5 Settlement of building MA0210

On MA0230, monitoring points are stable.

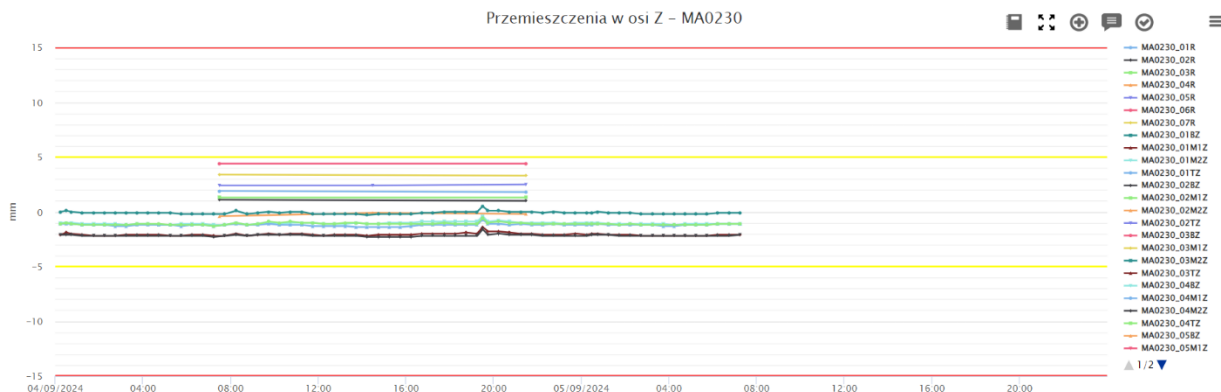


Figure 6-6 Settlement of building MA0230

7. COMPENSATION GROUTING ACTIVITIES

According to Raport Dobowy 04/09/2024 of Soletanche, compensation grouting has been performed on the 04/09/2024 from Shaft 3 in Próchnika 42.

The total amount of declared injected volume is 28722.56 l.

8. SOIL CONDITIONING

The volume of injected foam is plotted in the following Figure. The FIR value is computed based on the excavated material amount per ring and reported in the Table.

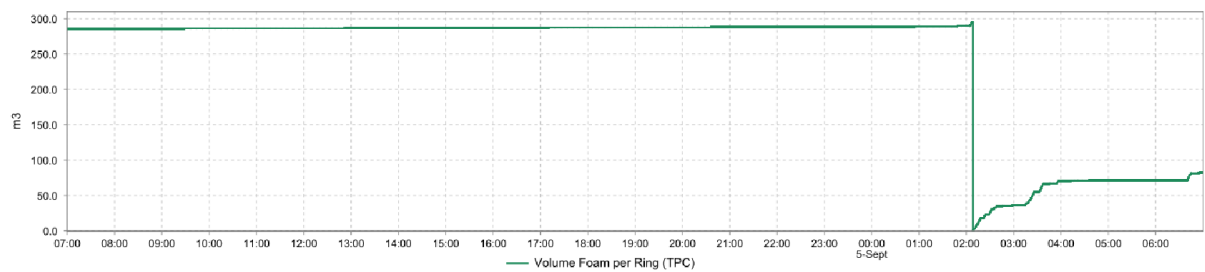


Figure 8-1 Injected foam volume

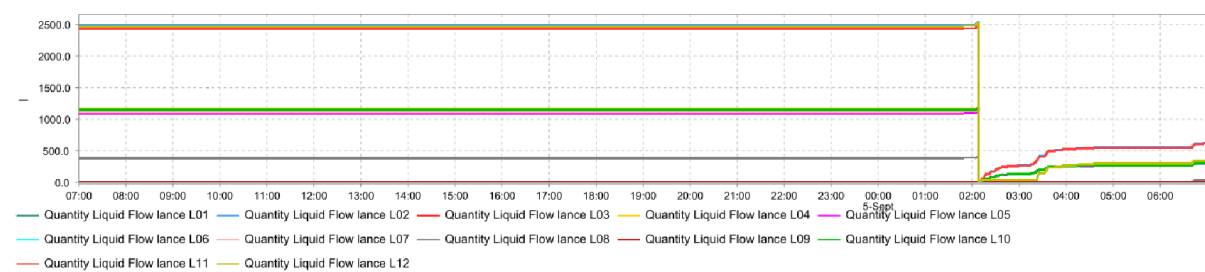


Figure 8-2 Conditioning liquid volume

Table 8-1 Soil conditioning data

Ring	Liquid volume (m³)	C (%)	FIR (%)
258	16	2	140